

# Same Mean, Different Deadline Risk

AP Statistics · Topic 4.8 (CED 2.9) · B: 2026-11-02 · E: 2026-11-23 · TEACHER KEY

## CED TETHER

- **Skill 3.B** — Determine parameters for probability distributions.
- **Skill 4.B** — Interpret statistical calculations and findings to assign meaning or assess a claim.
- **EU VAR-5** — Probability distributions may be used to model variation in populations.
- **LO VAR-5.C** — Calculate parameters for a discrete random variable. [Skill 3.B]
- **EK VAR-5.C.1** — A numerical value measuring a characteristic of a population or the distribution of a random variable is known as a parameter, which is a single, fixed value.
- **EK VAR-5.C.2** — The mean, or expected value, for a discrete random variable  $X$  is  $\mu_X = \sum x_i \cdot P(x_i)$ .
- **EK VAR-5.C.3** — The standard deviation for a discrete random variable  $X$  is  $\sigma_X = \sqrt{[\sum (x_i - \mu_X)^2 \cdot P(x_i)]}$ .
- **LO VAR-5.D** — Interpret parameters for a discrete random variable. [Skill 4.B]
- **EK VAR-5.D.1** — Parameters for a discrete random variable should be interpreted using appropriate units and within the context of a specific population.

**Essential question:** When two probability models have the same mean, how can standard deviation change the decision?

**DOK-3 skill:** 4.B · **FRQ pattern:** same-mean-different-sd-which-parameter-answers-the-question

**DOK rationale:** Part (c) weighs equal means against sharply different variability for a deadline-sensitive decision, carries both parameters through a transformation, and explains the practical consequence hidden by a mean-only comparison.

**Do Now (first take) → Explore (video + follow-along) → Exit (finish + turn in)** **DOK: 1**  
→ 3 **Minutes: 45**

### Teacher does

- Board slide up at the bell. Ask for a route choice before discussing parameter formulas.
- Begin the parameter portion of the combined 4.7–4.8 follow-along at minute 5.
- Return to the route models at minute 33. Circulate and connect spread to this customer’s deadline.
- Collect at the door. Score part (c) only, E/P/I.

### Students do

- Choose a route and cite one model feature.
- Complete the video follow-along about means, standard deviations, and their interpretations.
- Finish (a)–(c), revise the first take in the exit line, and turn in the sheet.

### Questions to ask

- What do the equal means say, and what do they leave unresolved?
- Which route places probability beyond the 40-minute deadline?
- When every outcome increases by five, what happens to each distance from its new mean?

- Would your decision be the same for a customer who valued the chance of arriving extremely early?

### Adult role

Solo teacher. Circulate 5 pairs. Require both a parameter and a deadline-specific display feature; do not let same average stand in for same reliability.

### [WARN] WATCH FOR

- Interpreting a standard deviation as the exact deviation of every delivery.
- Using sample standard deviation with denominator 7 instead of the probability-model denominator 8.
- Adding five to standard deviation when loading time is constant.

## SCORING PART (c) — E / P / I

### Expected elements

- **chooses-low-variability-route** (required) — Chooses Route A and cites its 3-minute standard deviation versus Route B's 12-minute standard deviation
- **connects-display-to-deadline** (required) — Uses Route A's 25-to-35 range and no outcomes above 40 versus Route B's two outcomes above 40
- **diagnoses-mean-only-claim** (required) — Explains that equal centers do not imply equal consistency or equal late risk
- **transforms-both-parameters** (required) — Adds 5 to both means, leaves both standard deviations unchanged, and reevaluates the decision

|          |                                                                                                                                                                                            |
|----------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>E</b> | Chooses Route A using both standard deviation and deadline evidence, explains the mean-only misconception, correctly transforms all four parameters, and confirms the post-shift decision. |
| <b>P</b> | Chooses Route A with relevant variability evidence but omits the practical mean-only consequence, one transformed parameter, or the revised deadline comparison.                           |
| <b>I</b> | Says the routes are interchangeable because their means match, adds 5 to either standard deviation, or chooses without connecting variability to lateness.                                 |

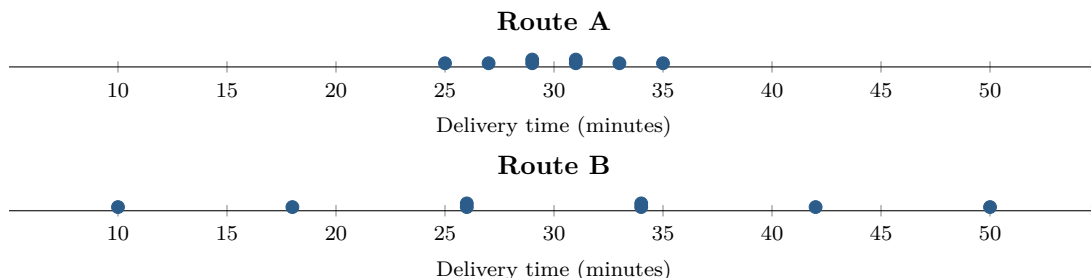
### Common mistakes

- Treating standard deviation as extra minutes added to every delivery
- Claiming every Route B delivery is 12 minutes from 30
- Adding 5 to the standard deviations when loading time is constant
- Using the equal means alone to conclude equal probabilities of missing 40 minutes

## [STAR] TODAY'S PROBLEM — annotated

Suppose a dispatcher models delivery time using eight equally likely outcomes for each route, shown in the dotplots. An analyst reports that both routes have mean 30 minutes, with  $\sigma_A = 3$  minutes and  $\sigma_B = 12$  minutes.

The dispatcher says, *The routes have the same mean, so it does not matter which one a customer gets.* A customer has 40 minutes before an appointment and places a high cost on being late.



### FIRST TAKE (5 min)

Which route would you assign to this customer? Cite one number or dotplot feature.

**Key:** Not graded. Route B's very fast outcomes may attract some students before deadline risk is made explicit.

*Rules callout "MEAN, STANDARD DEVIATION, AND SHIFTS (keep on the page)" is printed on the student sheet.*

**DOK 1 (a)** For each route, state its minimum and maximum modeled time and count the outcomes longer than 40 minutes.

**Key:** Route A has minimum 25, maximum 35, and 0/8 outcomes above 40 minutes. Route B has minimum 10, maximum 50, and 2/8 = 25% above 40: 42 and 50.

**DOK 2 (b)** Verify the reported mean and population standard deviation for each route from the eight equally likely outcomes. Show the sums of squared deviations.

**Key:** Both sets sum to 240, so each mean is  $240/8 = 30$  minutes. For Route A, the deviations are  $-5, -3, -1, 1, 1, 3, 5$  and the squared deviations sum to  $25 + 9 + 1 + 1 + 1 + 1 + 9 + 25 = 72$ , so  $\sigma_A = \sqrt{72/8} = 3$ . For Route B, deviations are  $-20, -12, -4, -4, 4, 4, 12, 20$  and their squares sum to  $400 + 144 + 16 + 16 + 16 + 16 + 144 + 400 = 1152$ , so  $\sigma_B = \sqrt{1152/8} = 12$  minutes.

**DOK 3 (c)** Decide which route better matches this customer's priority. Cite the parameter and dotplot feature that settle your choice, and explain what the dispatcher's mean-only argument would mislead a reader into believing. Then suppose every delivery gets 5 minutes of loading time: state both transformed means and standard deviations and decide whether your route choice changes.

**Key:** Choose Route A:  $\sigma_A = 3$  rather than  $\sigma_B = 12$ , and none of its times exceed 40, versus 2/8 = 25% for Route B. Equal means describe center only, so the dispatcher's claim conceals unequal reliability and deadline risk. Adding 5 makes both means 35 but leaves the standard deviations 3 and 12 because a constant shift preserves deviations. The new ranges are 30–40 and 15–55, so Route A remains the supported choice.

### EXIT

One thing the video changed about my first take: \_\_\_\_\_